# ASSSIGNMENT NO: 8

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| **PRN NO: 125B2F005** |
| **SUBJECT : DATABASE MANAGEMENT SYSTEM** |
| **DIV: IT A** |

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**Aim:**

**NoSQL: Aggregation and Indexing – Design and Develop Queries using Aggregation and Indexing with suitable example using MongoDB**

**Theory**

**1. Aggregation in MongoDB**

Aggregation is used to process multiple documents and return computed results.
It works like SQL `GROUP BY`, `SUM`, `AVG`, `COUNT`, etc.

**Common Aggregation Stages:**

- `$match` → Filters documents
- `$group` → Groups data
- `$sum` → Calculates total
- `$avg` → Calculates average
- `$sort` → Sorts results
- `$project` → Displays selected fields

**Advantages:**

- Fast data analysis
- Performs calculations on grouped records
- Useful for reports and summaries

**2. Indexing in MongoDB**

Indexing improves query performance by reducing search time.

Without indexing:
→ MongoDB scans entire collection

With indexing:
→ MongoDB directly locates required documents

**Types of Indexes:**

1. Single Field Index
2. Compound Index
3. Unique Index
4. Text Index

**Advantages:**

- Faster searching
- Better sorting performance
- Optimized query execution

MongoDB uses `createIndex()` to create indexes.

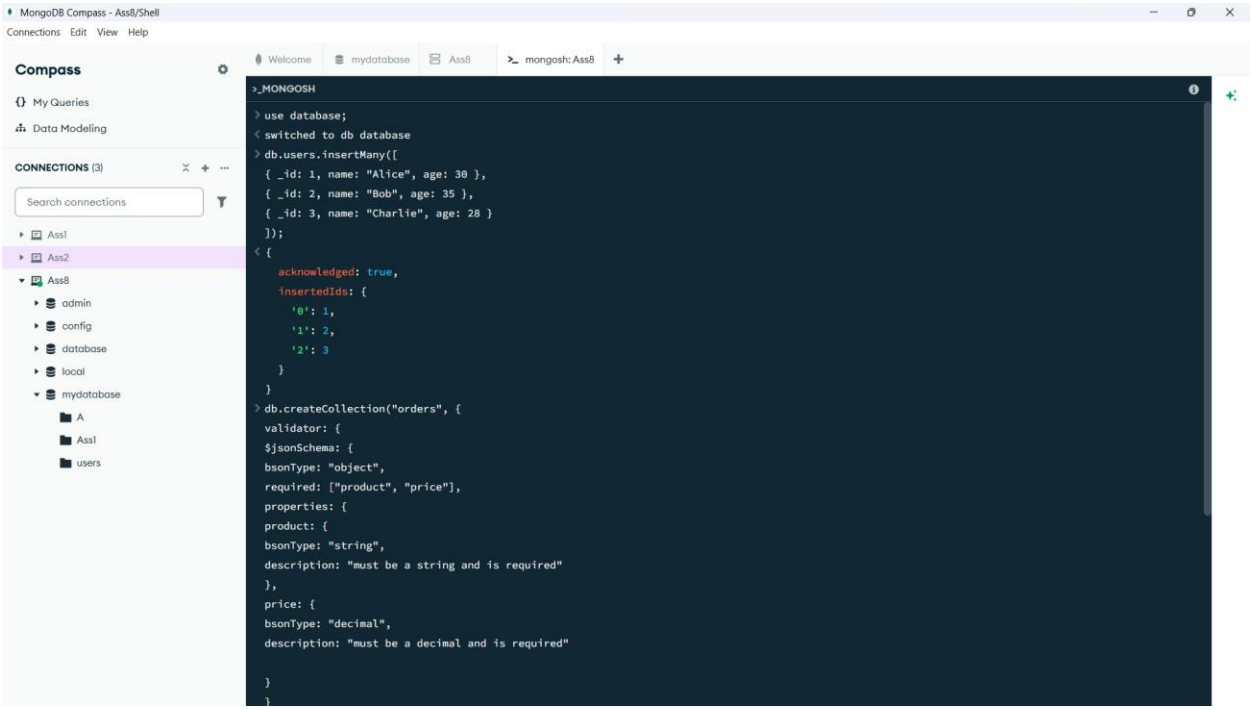
## MONGODB

**Step1:** install using below link--->https://www.mongodb.com/try/download/community

**Step2:** add environment variable upto bin folder

**Step3:** Create New Connection bydefault give Name

**Step4:** execute below commands




use mydatabase;

switched to db mydatabase

```
db.users.insertMany([
{ _id: 1, name: "Alice", age: 30 },
{ _id: 2, name: "Bob", age: 35 },
{ _id: 3, name: "Charlie", age: 28 }
]);
{
acknowledged: true,
insertedIds: {
'0': 1,
'1': 2,
'2': 3
}
}
db.createCollection("orders", {
validator: {
$jsonSchema: {
bsonType: "object",
required: ["product", "price"],
properties: {
product: {
bsonType: "string",
description: "must be a string and is required"
},
price: {
bsonType: "decimal",
description: "must be a decimal and is required"

}
}
}
```

```
}
});
{ ok: 1 }
db.orders.insertOne({ price: 50.0 });
MongoServerError: Document failed validation
db.users.remove({ name: "Charlie"});
DeprecationWarning: Collection.remove() is deprecated. Use deleteOne, deleteMany, findOneAndDelete,
or bulkWrite.
{
acknowledged: true,
deletedCount: 1
}
db.users.remove({});
{
acknowledged: true,
deletedCount: 2
}
db.users.update(
{ _id: 1 },
{ $set: { name: &quot;Alicia&quot;, age: 31 } }
);
SyntaxError: Unexpected token (3:16)

1 | db.users.update(
2 | { _id: 1 },
> 3 | { $set: { name: &quot;Alicia&quot;, age: 31 } }
| ^
4 | );
db.users.update(
{ _id: 1 },
```

```
{ $set: { name: "Alicia", age: 31 } }
);

DeprecationWarning: Collection.update() is deprecated. Use updateOne, updateMany, or bulkWrite.
{
acknowledged: true,
insertedId: null,
matchedCount: 0,
modifiedCount: 0,
upsertedCount: 0
}
db.users.update({ _id: 2 }, { $inc: { age: 1 } });
{
acknowledged: true,
insertedId: null,
matchedCount: 0,
modifiedCount: 0,
upsertedCount: 0
}
db.users.update(
{ _id: 4 },
{ $set: { name: "Eva", age: 29 } },
{ upsert: true }
);
{
acknowledged: true,
insertedId: 4,
matchedCount: 0,
modifiedCount: 0,
upsertedCount: 1
```

```
}
db.users.update({ age: { $gt: 30 } }, { $set: { status: "Senior" } }, { multi: true });
{
acknowledged: true,
insertedId: null,
matchedCount: 0,
modifiedCount: 0,
upsertedCount: 0


}
db.users.find({});
{
_id: 4,
age: 29,
name: 'Eva'
}
db.users.insertMany([
{ _id: 1, name: "Nikita", age: 30 },
{ _id: 2, name: "Vedant", age: 35 },
{ _id: 3, name: "Rushi", age: 28 }
]);
{
acknowledged: true,
insertedIds: {
'0': 1,
'1': 2,
'2': 3
}
}
mydatabase
```

Selection deleted

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## Conclusion:

This assignment demonstrates MongoDB Aggregation and Indexing successfully.
Aggregation helps analyze grouped data efficiently, while indexing improves query speed and performance.
Both are essential for handling large datasets in NoSQL databases.

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